June 30, 1925. 1,543,782

F. W. MAUE

REED SHIFTING MEANS FOR WARPING MACHINES

Filed Feb. 2, 1924 5 Sheets-Sheet 1

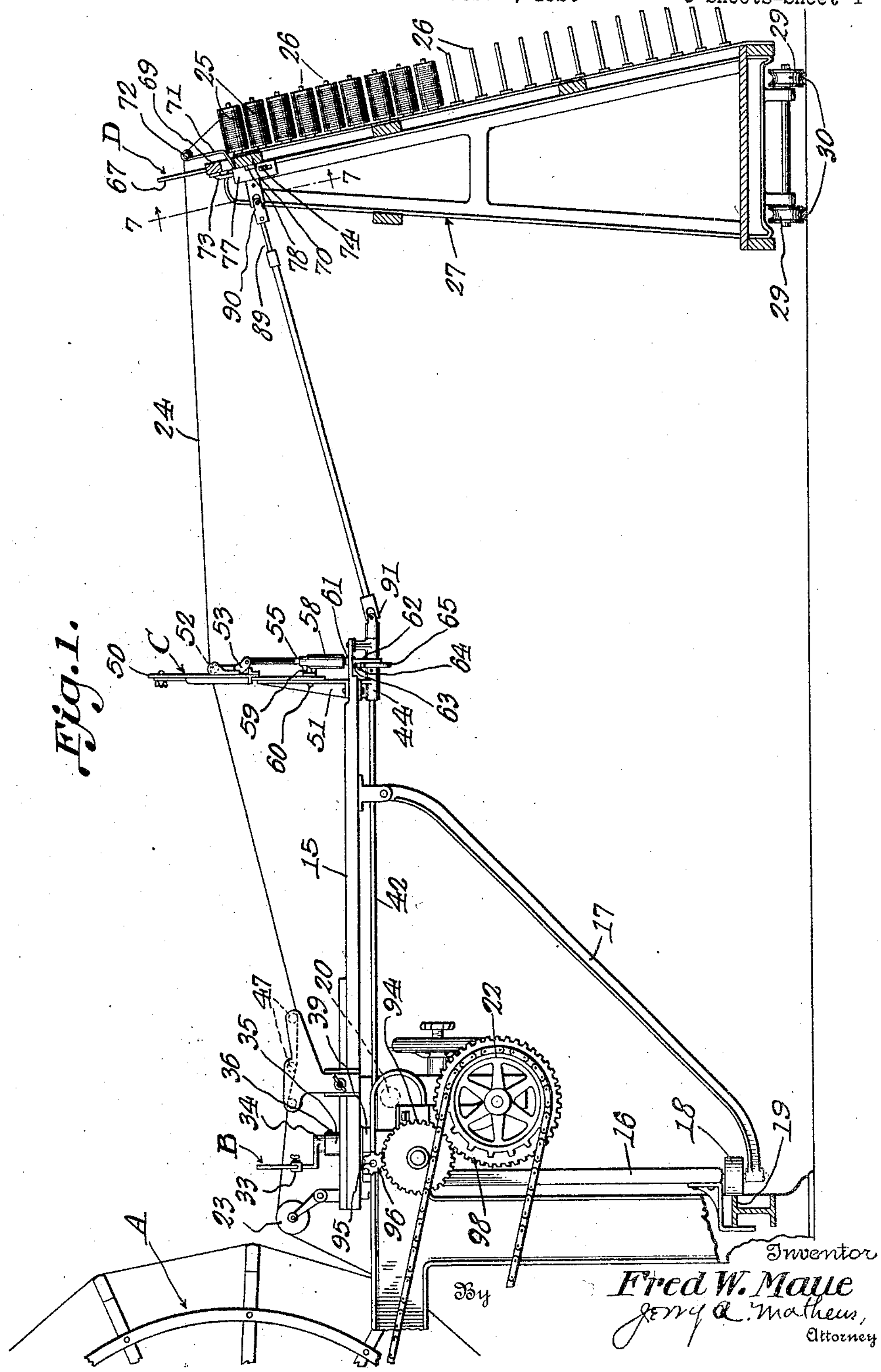

Fig. 1.

Inventor

Fred W. Maue

By Jerry A. Mathews, Attorney

June 30, 1925.

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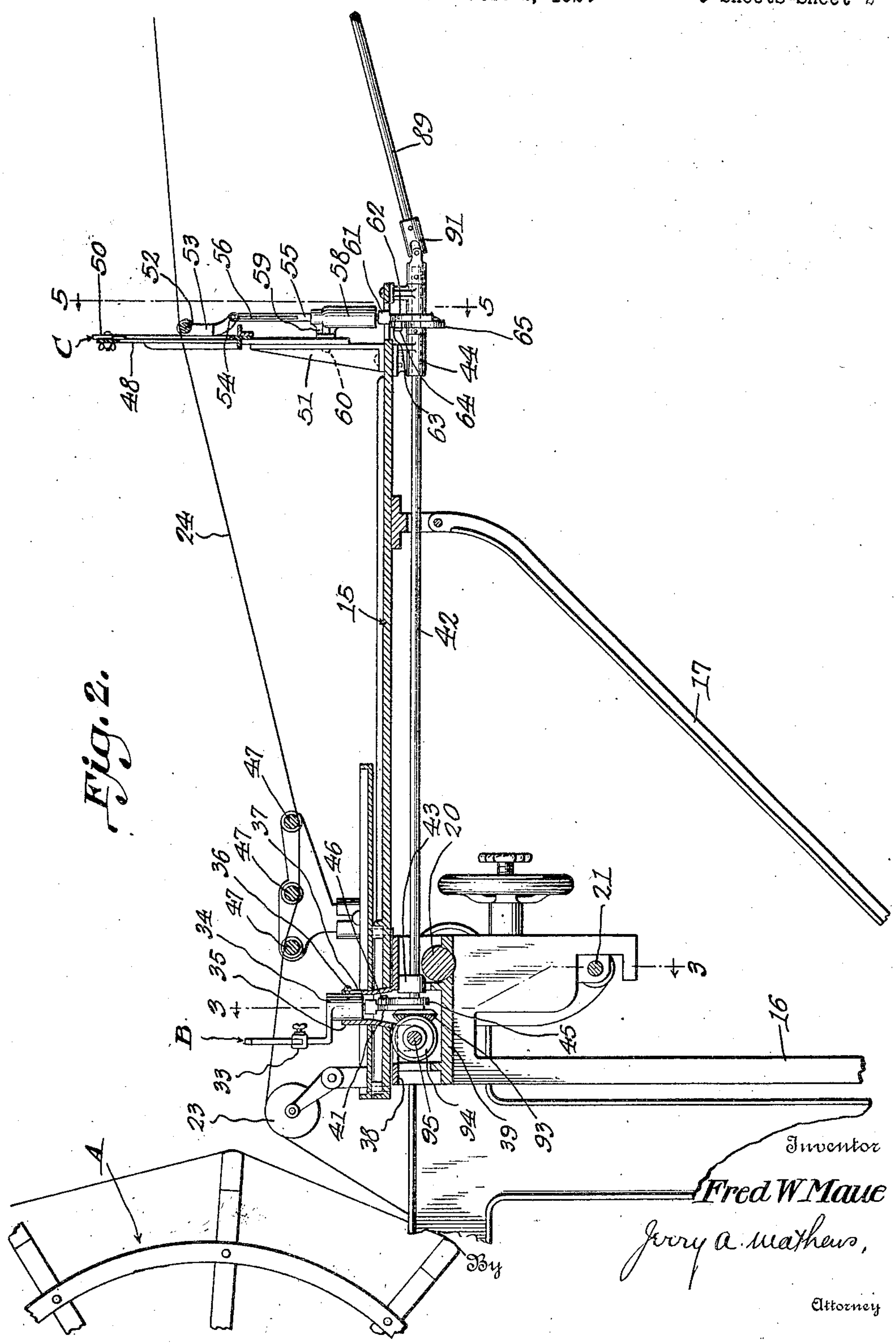
5 Sheets-Sheet 2



Inventor

Fred W. Maue

By Jerry A. Mathews,

Attorney

June 30, 1925.

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5 Sheets-Sheet 3

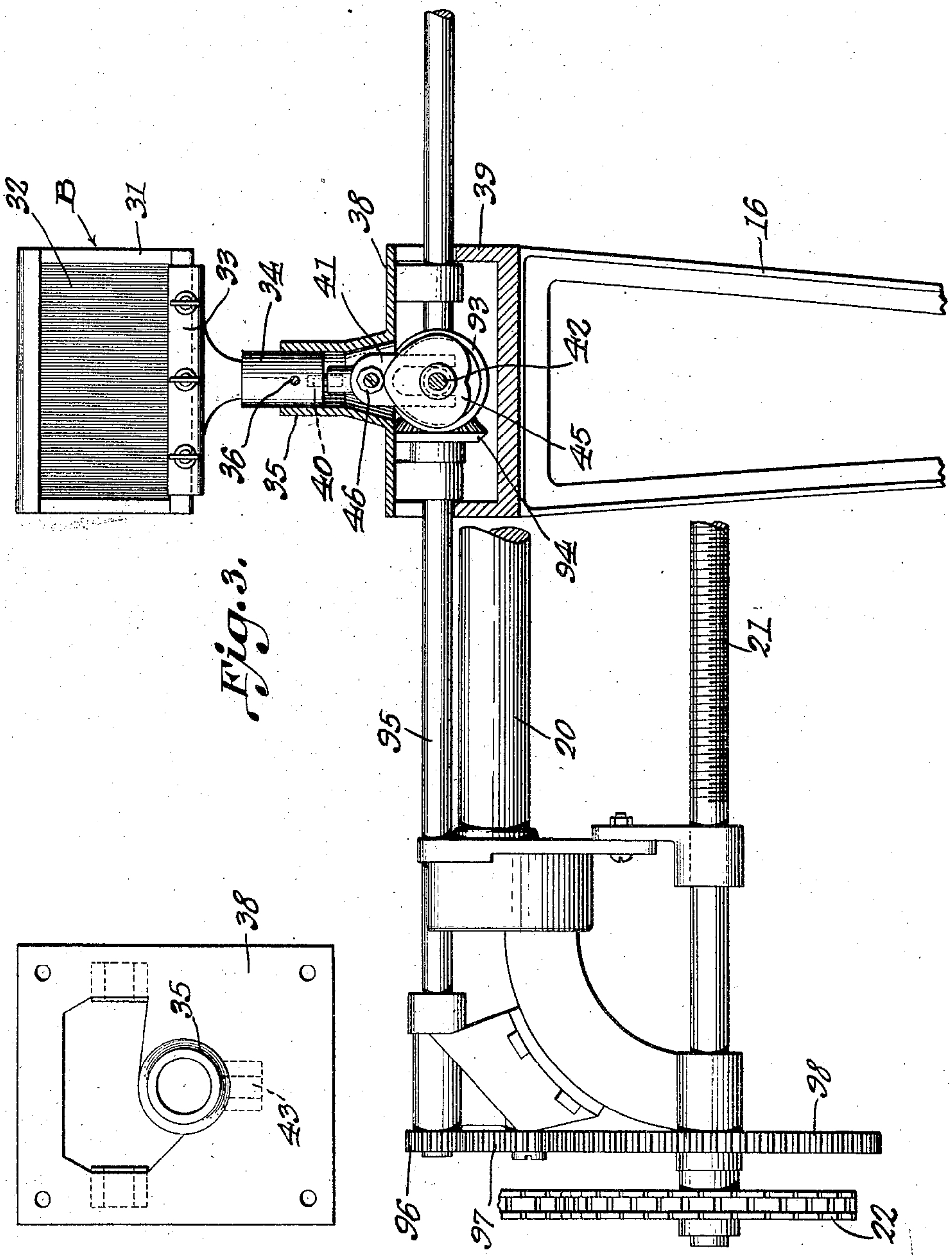

Fig. 3.

Fig. 4.

Inventor

Fred W. Maue

By Jerry A. Mathews,

Attorney

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F. W. MAUE

REED SHIFTING MEANS FOR WARPING MACHINES

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Fig. 6.

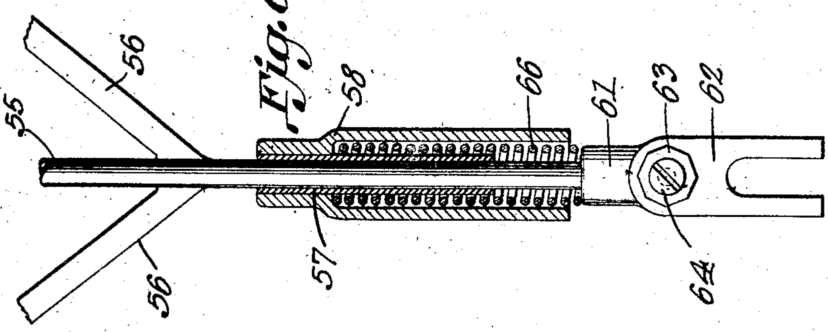
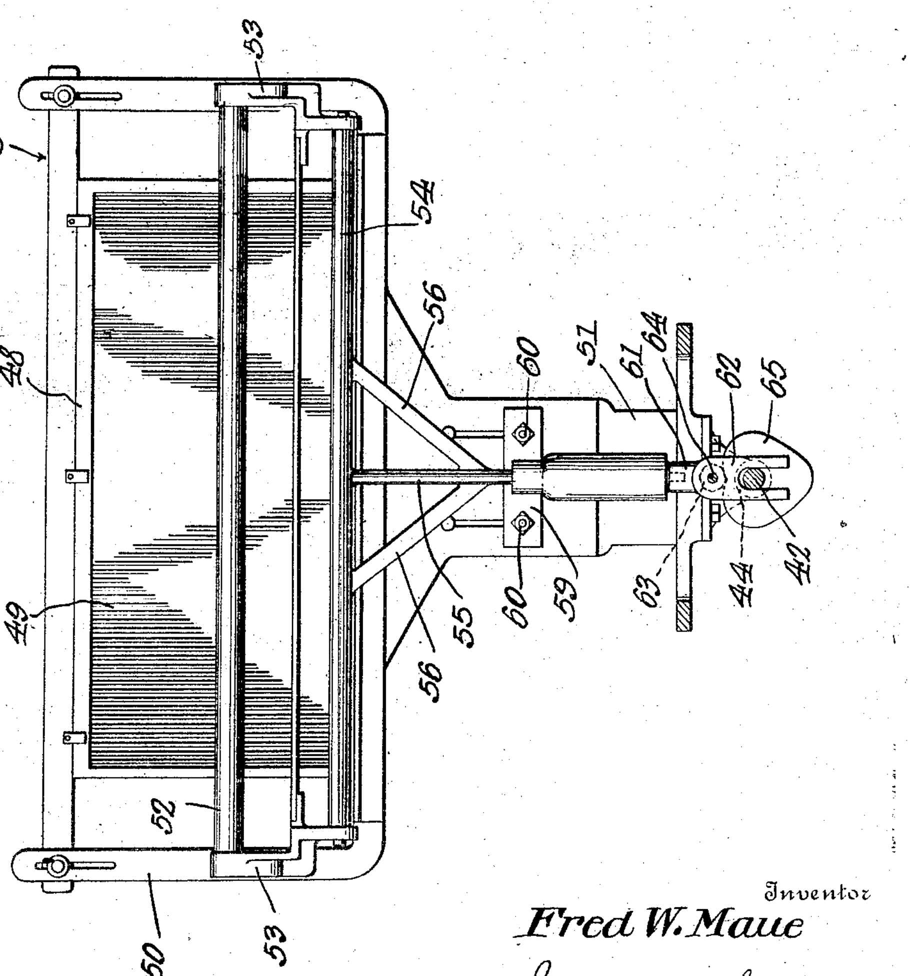


Fig. 5.



Inventor

Fred W. Maue

By Irving A. Mathews.

Attorney

June 30, 1925. 1,543,782

F. W. MAUE

REED SHIFTING MEANS FOR WARPING MACHINES

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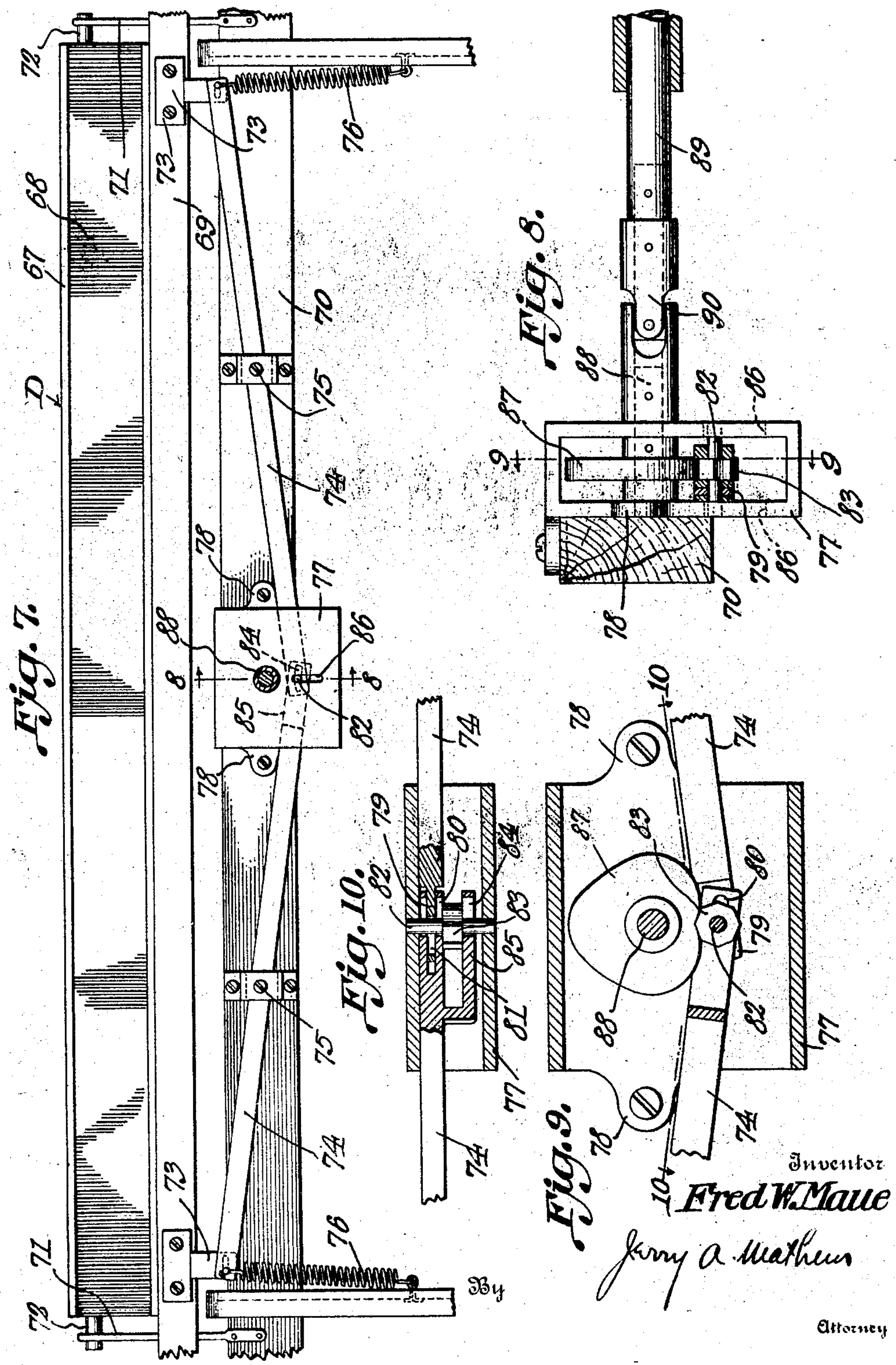

Patented June 30, 1925. **1,543,782**

UNITED STATES PATENT OFFICE.

FREDERICK WILLIAM MAUE, OF SHAMOKIN, PENNSYLVANIA.

REED-SHIFTING MEANS FOR WARPING MACHINES.

Application filed February 2, 1924. Serial No. 690,272.

To all whom it may concern:

Be it known that I, FREDERICK WILLIAM MAUE, a citizen of the United States, residing at Shamokin, in the county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Reed-Shifting Means for Warping Machines, of which the following is a specification.

My invention relates to means for effecting a relative lateral movement between warp threads and reeds, in the operation of a warping mill.

In the operation of a warping mill, for the production of warps, made from various threads, it is necessary that the threads or yarn be drawn from bobbins or spools carried on a creel. This creel consists of a frame work with spindles or pins, carrying the bobbin or spools. The thread from these bobbins or spools is ordinarily passed through a number of reeds, such as the creel reed, cross reed, and the section reed. As a result of the frictional engagement of the thread with the sides of the dent wires of these reed or reeds, such dent wires become worn or cut, at the point of contact, and it is necessary to frequently replace the reeds with new reeds, for after the dent wires become cut, they cause the threads to rub and chafe, whereby many of them break.

An important object of the invention is to provide means of the above mentioned character, whereby this relative movement is effected mechanically and continuously during the operation of the warping mill, so that the wear frequently occurring upon the dent wires is eliminated.

A further object of the invention is to provide means of the above mentioned character, which is wholly automatic in operation.

A further object of the invention is to provide means of the above mentioned character, which may be installed upon a warping mill without materially altering the construction thereof.

Other objects and advantages of the invention will be apparent during the course of the following description.

In the accompanying drawings forming a part of this specification, and in which like numerals are employed to designate like parts throughout the same,

Figure 1 is a side elevation of apparatus embodying my invention,

Figure 2 is a central vertical longitudinal section through the same,

Figure 3 is a transverse vertical section taken on line 3—3 of Figure 2,

Figure 4 is a plan view of a base element,

Figure 5 is a transverse section taken on line 5—5 of Figure 2,

Figure 6 is a detail section through the guide sleeve of the cross reed,

Figure 7 is a transverse section taken on line 7—7 of Figure 1,

Figure 8 is a vertical section taken on line 8—8 of Figure 7,

Figure 9 is a vertical section taken on line 9—9 of Figure 8, and,

Figure 10 is a horizontal section on line 10—10 of Figure 9.

In the drawings, wherein for the purpose of illustration is shown a preferred embodiment of my invention, A designates the revoluble reel of a warping mill, B the section reed, C the cross reed, and D the creel reed.

The warping machine embodies a carriage 15, provided with a vertical leg 16 and a diagonal leg 17. The leg 16 carries a roller 18, traveling upon a track 19. The carriage is slidable upon a horizontal transverse rod 20, and this carriage is shifted longitudinally of the reel A by means of a feeding screw 21, driven from the sprocket wheel 22 of the warping mill. The carriage 15 is equipped with a guide roller 23, over which the warp thread passes prior to its winding upon the reel of the warping mill. The carriage 15 is advanced longitudinally of the reel, as above stated, to properly apply the wrap thread thereto. This is the ordinary construction of the well known warping machine, and it is thought to be unnecessary to further explain the same.

The warp threads 24 are unwound from spools or bobbins 25, rotatable upon pins 26, carried by creel 27. This creel is mounted upon grooved wheels 29, traveling on tracks 30. The creel may be shifted to a proper position upon the track, so that it will be properly located with respect to the reel of the warping machine.

The section reed B, see more particularly Figure 3, embodies a reed frame 31, carrying dent wires 32, through which the warp threads 24 pass. The frame 31 is held within a clamp 33 carried by a guide plunger 34. This guide plunger 34, see Figures 2 and 3, reciprocates within a vertical guide sleeve 35, having a screw 36 attached thereto, operating within a vertical slot 37 in the guide sleeve. The guide sleeve 35 is preferably integral with a base 38, which is attached to the top of a guide casing or box 39. Rigidly attached to the lower end of the guide plunger 34 is a coupling rod 40, which is rigidly connected with a forked shifting member 41, which straddles a longitudinal power shaft 42, journaled in bearings 43 and 44. This power shaft carries a heart shaped cam 45, rigidly mounted thereon, operating at the side of the forked member 41, and contacting with a thrust element or nut 46, which is clamped to the forked member 41 by means of a screw or the like. The thrust member 46 is preferably polygonal so that it may be turned upon its screw and clamped to the forked member 41 so that an unworn edge may be presented to contact with the cam 45. The section reed B is lowered by gravity, and raised by the action of the cam 45 and associated elements, and it is thus seen that as the power shaft 42 is continuously rotated, this section reed will be reciprocated up and down. The warp threads 24 pass about cylinders or rods 47, which are preferably formed of glass, which serve to guide them, prior to their passage between the dent wires of the section reed B.

The cross reed C, see more particularly Figures 1, 2, 5 and 6, embodies a stationary reed frame 48, carrying dent wires 49. The reed frame 48 is held within an outer stationary frame 50, which is bolted or clamped to a stationary bracket 51, rigidly mounted upon the rear end of the table 15. The warp threads 24 are moved or reciprocated vertically, with respect to the stationary cross reed C, and this is accomplished by means of a vertically reciprocating warp lifting element including a rod 52, preferably formed of glass, and mounted in end members 53, which are rigidly mounted upon a tube 54, in turn rigidly mounted upon a vertical rod 55 having diagonal braces 56, as shown. The rod 55 is slidable within a sleeve or bushing 57 held in the contracted end of a vertical guide sleeve 58, see more particularly Figure 6. This guide sleeve 58, Figures 1 and 2, has a laterally extending bracket 59, bolted to the bracket 51, as indicated at 60. The rod 55 has a block 61, rigidly mounted upon its lower end, carrying a forked head 62, which straddles the power shaft 42. A thrust element 63, polygonal in shape, is mounted upon the forked head 62 by means of a screw 64, or the like, and may be clamped to this forked head in a selected position, for causing one of its edges to contact with a heart shaped cam 65, rigidly mounted upon the power shaft 42. This cam serves to raise the rod 55, and this rod is forced downwardly by a compressible coil spring 66, mounted within the enlarged portion of the sleeve 58 and engaging the block 61. The warp threads pass over and engage with the glass rod 52, as is obvious.

In connection with the creel reed D, attention is called more particularly to Figures 1, 7, 8, 9 and 10. In these Figures, the reel D embodies a frame 67, carrying the dent wires 68. This frame is mounted upon a vertically movable or reciprocatory horizontal bar 69, mounted upon the creel frame, in a manner to be described. The numeral 70 designates the upper horizontal stationary beam of the creel frame, and upstanding arms 71 are mounted upon this beam rearwardly of the reel D, and carry a rod 72, preferably of glass, about which the warp threads 24 pass. The reed D is supported by the reciprocatory bar 69, as above explained, the ends of which operate within suitable guides to cause the same to partake of up and down movement without twisting or lateral movement. The bar 69 carries brackets 73, to which are pivotally connected by means of a slot and pin joint, vertically swinging levers 74, pivoted to the creel frame beam 70, as shown at 75. The outer ends of these levers 74 are pulled downwardly by means of retractile coil springs 76. The inner end of the levers 74 project into a housing or box 77, having lugs or ears 78, which are screwed or otherwise rigidly attached to the beam 70. As more clearly shown in Figure 10, within the box 77, the lever 74 to the right, has a tongue 79, and the lever 74 to the left has its end forked for receiving this tongue. The forked end has longitudinal slots 80 and the tongue has a longitudinal slot 81. These slots receive a pin 82, carrying a trip element or nut 83, rigidly attached thereto. The pin 82 also extends through a slot 84 in the arm 85, integral with the inner end of the lever 74 to the left. The pin 82 operates within vertical slots 86, formed in the box 77. The depression element or nut 83 is arranged beneath a heart shaped cam 87, rigidly mounted upon a stub shaft 88, carried by the box 77, above the depression element 83. This stub shaft is free to turn with relation to this box but cannot move longitudinally with respect thereto.

The stub shaft 88 is connected with a coupling shaft 89 by means of a universal joint 90, and the forward end of the shaft 89 is connected with the rear end of the power shaft 42, by means of a universal joint 91. It is thus seen that a flexible driving connection is had between the stub shaft 88 and the power shaft, whereby this stub shaft may be driven by the power shaft, when the creel frame is shifted longitudinally upon its track. The flexibility of the coupling shaft 89 embraces longitudinal adjustability so that this operation may occur.

The power shaft 42 is driven by a bevel gear 93, rigidly secured thereto, and this bevel gear is driven by a bevel gear 94, which is splined upon a transverse shaft 95. The gear 94 is arranged within the box 39 and is retained in permanent engagement with the gear 93. This gear 94 travels longitudinally of the shaft 95 but turns therewith. The shaft 95 carries a gear 96, engaging a gear 97, driven by a gear 98, which is turned by the sprocket wheel 22.

In the operation of the apparatus, the reel A of the warping mill is rotated in a direction to wind the warp thread 24 thereon, which warp thread passes through the several reeds. When passing through the section reed B, the reed is continuously reciprocated, vertically, and hence the wear of the dent wires is prevented. When passing through the cross reed C, the glass rod 52 is reciprocated vertically and hence the warp threads are continuously raised and lowered while traveling in contact with the dent wires of the reed, whereby wearing of the dent wires is prevented. When passing through the creel reed D, the reed is vertically reciprocated, and hence the wear of the dent wires is prevented.

It is to be understood that the form of my invention herewith shown and described, is to be taken as a preferred embodiment of the same, and that various changes in the shape, size, and arrangement of parts, may be resorted to, without departing from the spirit of my invention or the scope of the subjoined claims.

Having thus described my invention, I claim:

1. The combination with a creel embodying a frame, of a transversely reciprocatory reed mounted upon said frame, a pair of levers pivoted between their ends to the frame and having pivotal connection at their outer ends with the reed, means to move the outer ends of the levers downwardly, a pin pivotally connecting the inner ends of the levers, a stop element carried by the pin, and a cam engaging the stop element to shift the same laterally.

2. The combination with a creel embodying a frame, of a laterally reciprocatory reed mounted upon the frame, a pair of levers pivoted between their ends to the frame and having their outer ends pivotally connected with the reed, means to move the outer ends of the levers downwardly, a housing having a guide slot, a pin operating within the guide slot and pivotally connecting the inner ends of the levers, an element carried by the pin, and a cam operating in engagement with the element to depress the same.

3. A warping machine embodying a revoluble reel, a receiprocatory reed arranged upon one side of the reel, a stationary reed in advance of the reciprocatory reed, a reciprocatory thread shifting element arranged adjacent to the stationary reed, a power shaft, means operated by the power shaft to effect the reciprocatory movement of the reciprocatory reed, means operated by the power shaft to effect the reciprocatory movement of the thread shifting element, a reciprocatory reed to be mounted upon the movable creel, means for effecting the reciprocatory movement of the last named reed, and a flexible connection between the power shaft and the last named means.

4. A warping machine embodying a revoluble reel, a carriage to travel longitudinally of the reel, a power shaft extending longitudinally of the carriage, means to drive the power shaft, a reciprocatory section reed mounted upon the carriage, means driven by the power shaft for effecting the movement of the section reed, a stationary cross reed mounted upon the carriage, a reciprocatory thread shifting element arranged adjacent to the cross reed, means driven by the power shaft to move the reciprocatory thread shifting element, a creel movable with relation to the carriage, a reciprocatory creel reed mounted upon the creel, means to effect the movement of the creel reed, and a flexible connection between the power shaft and last named means.

5. The combination with a creel embodying a frame, of a vertical movable reed mounted upon the frame, vertically swinging levers pivoted between their ends upon the frame and having their outer ends connected with the reed, means to draw the outer ends of the levers downwardly, a support, a cam mounted upon the support, and an element connecting the inner ends of the levers and adapted to be depressed by the cam.

6. The combination with a creel embodying a frame, of a vertically movable reed mounted upon the frame, vertically swinging levers pivoted between their ends with the frame and having their outer ends connected with the reed, a support having a substantially vertical slot, a pin connecting the inner ends of the levers and operating within the slot to be guided thereby, and a substantially heart shaped cam mounted upon the support above the pin to depress the same.

7. The combination with a warping machine embodying a revoluble reel and a carriage to travel longitudinally of the reel, a section reed mounted upon the carriage, a cross reed mounted upon the carriage, means for effecting a relative movement between

said reeds and the warp threads embodying a power shaft extending longitudinally of the carriage and driven from a movable part of the warping machine, a creel arranged near the warping machine, and movable bodily with relation thereto, a creel reed carried by said creel, and means for effecting a relative movement between the creel reed and the warp threads embodying a flexible coupling shaft connected with said power shaft and driven thereby.

In testimony whereof I affix my signature.

FREDERICK WILLIAM MAUE.